Attractor Geometry of Life:

Pre-Registration of Geometric Derivations, Experimental Predictions, and Falsifiability Conditions

Covering the Reproduction Theorem, the Inverse Accessibility Theorem, the LECA Resurrection Protocol, the Dickinsonia Protocol, the Cross-Kingdom Convergence Prediction, and the Geometric Status of the LUCA

Derived from Waddington Landscape Attractor Geometry and the Triadic Invariant
OrganismCore Document: ATTRACTOR-LIFE-PREREGISTRATION-v1.0

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Repository: github.com/Eric-Robert-Lawson/attractor-oncology

Pre-registration date: 2026-03-12

All predictions in this document are pre-specified prior to any experimental confirmation. This document constitutes the formal pre-registration record.

Full derivation record, protocols, reasoning artifacts, and pre-registration documents:

https://github.com/Eric-Robert-Lawson/attractor-oncology

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Topics: attractor · bioinformatics · cancer-genomics · computational-oncology · drug-discovery · epigenetics · evolutionary-biology · developmental-biology · de-extinction

All derivations are timestamped and version-controlled. All predictions are pre-specified before experimental confirmation.

Abstract

We present the formal pre-registration record for a set of geometric derivations, theorems, and experimentally testable predictions produced by the Attractor Geometry framework — a principles-first geometric framework derived from the triadic invariant as applied to biological identity at all scales. The framework was originally developed to derive cancer drug targets from epigenetic geometry and has been extended to evolutionary biology, developmental biology, and synthetic organism construction.

The central result is the **Reproduction Theorem**: the developmental trajectory of any organism from informational encoding to adult form is geometrically identical to the evolutionary trajectory of its lineage, compressed into developmental time. This is established by proof of elimination: no alternative coherent trajectory from

informational biological encoding to a living organism has been described in 150 years of developmental biology.

The theorem generates the **Inverse Accessibility Theorem**: the oldest recoverable organism is the easiest and cheapest to recover, because evolutionary age is inversely proportional to developmental position, and the earlier the developmental position the simpler the arrest intervention.

The most accessible experimental consequence is the **LECA Resurrection Protocol**: developmental arrest of any eukaryotic cell before its first multicellular organisational step will produce an organism morphologically consistent with the Last Eukaryotic Common Ancestor (∼1.5 Ga). This result is predicted to be morphologically identical whether the starting material is an animal embryo, a plant seed, or a fungal spore — the **Cross-Kingdom Convergence Prediction** — while being genomically distinct under sequencing. The experiment costs $200–400 per run and is executable in any standard cell biology laboratory in days.

We further establish the geometric status of the LUCA: the ribosome's peptidyl transferase center constitutes the continuously executing Row 3 identity of the Last Universal Common Ancestor. The LUCA is not extinct by the precise definition of the triadic invariant. The ribosome is identified as the topological origin of the structural space of all life and as the medium on which every developmental arrest runs — present inside every recovered ancestral organism, older than every attractor it will ever be found in.

All predictions are pre-specified with explicit falsifiability conditions. Pre-registration timestamp: 2026-03-12.

Keywords: attractor geometry; triadic invariant; Reproduction Theorem; Inverse Accessibility Theorem; LECA; LUCA; ribosome; developmental arrest; de-extinction; Dickinsonia; Ediacaran; cross-kingdom convergence; Waddington landscape; OrganismCore

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1 Document Metadata

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2 Preamble: The Derivation Path

This document records geometric derivations produced by Eric Robert Lawson, an independent autodidactic researcher with no institutional affiliation, no laboratory access, and no formal training in biology, working from first principles using the triadic invariant as the foundational structure.

The derivation path was not planned. It followed geometrically:

1. Cancer drug target geometry → the Identity Axis Theorem: the ratio $R = A/H$ (FOXA1/EZH2) as the epigenetic identity axis of committed breast epithelial cell types. Validated across ∼7,500 patients in seven independent datasets on four measurement platforms (see FOXA1-EZH2-RATIO-MASTER-RA, 2026-03-06).
2. Identity axis geometry → the triadic invariant as the causal structure of all biological identity at every scale.
3. Triadic invariant applied to organism persistence across time → the Reproduction Theorem.
4. Reproduction Theorem → Inverse Accessibility Theorem, LECA protocol, Dickinsonia protocol, plant Ediacaran protocol.
5. Ancestral access map → geometric status of the LUCA: the ribosome as the LUCA's continuously executing Row 3 identity. The LUCA is not extinct.
6. Ribosome as LUCA → the ribosome as topological origin of the structural space of all life and as the medium on which every developmental arrest runs.

The derivation of the ribosome's geometric role was produced without prior knowledge of ribosome structure or function. The structure was identified as the ribosome only after the

geometric role was derived. The literature confirmed the observation post-derivation. This epistemic pattern — geometry first, identification second, literature confirmation third — is consistent across all results in this framework.

3 The Triadic Invariant: Foundational Structure

Definition 3.1 (The Triadic Invariant)**.** All stable biological identity at every scale is constituted by a three-row structure:

- **Row 1** (Informational Basis): The substrate encoding the attractor geometry — the compressed record of evolutionary navigation.
- **Row 2** (Gap Navigation): The process by which Row 1 is decompressed into a physical traversal of the attractor landscape.
- **Row 3** (Stable Product): The coherent, persistent, self-referential identity that is the stable endpoint of Row 2 navigation and that maintains the conditions of its own production.

Definition 3.2 (Extinction)**.** An organism is extinct when its Row 3 product — its coherent, persistent, self-maintaining identity — ceases to be produced.

Definition 3.3 (Attractor)**.** An attractor is a stable configuration in the epigenetic or developmental landscape that Row 2 navigation reaches and maintains. Attractors are sequentially dependent: each attractor requires the stable product of the preceding attractor as its Row 1 substrate.

3.1 Prior validation of the triadic invariant

The triadic invariant was first validated in the cancer framework. Applied to breast cancer, it produced the FOXA1/EZH2 ratio as the epigenetic identity axis — the continuous measure of the balance between luminal identity maintenance (FOXA1) and identity suppression (EZH2). This prediction was locked before confirmatory analysis and confirmed across ∼7,500 patients in seven independent datasets with zero biological contradictions (see repository: `Cancer_Research/BRCA/DEEP_DIVE/MAJOR_FINDINGS/FOXA1_EZH2_RATIO/FOXA1_EZH2_BCRA_Ratio.tex`).

The same invariant, applied to the problem of organism persistence across evolutionary time, generates all results in this document.

4 The Reproduction Theorem

4.1 Statement

Theorem 4.1 (The Reproduction Theorem)**.** *The developmental trajectory of any organism from informational encoding (fertilised egg or germinating seed) to adult organism is geometrically identical to the evolutionary trajectory of its lineage, expressed at a compressed temporal scale.*

They are not analogous. They are not parallel. They are the same sequence of attractor states, in the same order, from the same informational starting point, to the same terminal state.

4.2 The developmental-evolutionary correspondence

Developmental Stage	Evolutionary Grade	Age (Ma)
Fertilised egg / first cleavages	First multicellular aggregates	∼800
Blastula	Colonial multicellular, earliest animal grade	∼700
Gastrula	Diploblast / early bilaterian	∼560–600
Neurula	Urbilaterian / early deuterostome	∼540–560
Pharyngeal arch stage (phylotypic)	Vertebrate ancestor	∼500
Organogenesis	Specific vertebrate lineage grades	∼400–300
Late fetal	Mammalian ancestor	∼200
Birth / adult	Species-specific	Present

This correspondence is confirmed by the developmental hourglass model across all major animal phyla [1, 2] and extended to plants [3].

4.3 Proof by elimination of alternatives

Proof. Let $\mathcal{T}_D$ denote the developmental trajectory and $\mathcal{T}_E$ the evolutionary trajectory of lineage L.

Premise 1. There exists exactly one known trajectory $\mathcal{T}_D$ from informational encoding to a living organism of any species. No alternative trajectory has been observed or demonstrated.

Premise 2. The stages of $\mathcal{T}_D$ are *sequentially dependent attractors*: each stage A_i is only accessible from A_{i-1}. Disruption of any stage is lethal. The order is geometrically necessary, not conventional.

Premise 3. The stages of $\mathcal{T}_D$ correspond stage-by-stage to the grades occupied by the lineage during $\mathcal{T}_E$, from earliest to most recently derived (confirmed by the developmental hourglass across all major animal and plant phyla).

Claim. $\mathcal{T}_D = \mathcal{T}_E$ (compressed).

Proof. Suppose $\mathcal{T}_D \neq \mathcal{T}_E$. Then one of the following must hold:

Alt. A. $\mathcal{T}_D$ skips one or more attractor states in $\mathcal{T}_E$. Excluded by Premise 2: skipping any attractor is lethal.

Alt. B. $\mathcal{T}_D$ traverses attractor states in a different order from $\mathcal{T}_E$. Excluded by sequential dependence: each stage's molecular preconditions are established only by the preceding stage in the observed order.

Alt. C. $\mathcal{T}_D$ traverses different attractor states and still reaches the same terminal organism. Requires the terminal attractor to be reachable from a sequence no lineage ever traversed. Not demonstrated in 150 years of developmental biology.

All alternatives are excluded. Therefore $\mathcal{T}_D = \mathcal{T}_E$ (compressed). □ □

4.4 Corollaries

Corollary 4.2. *Every reproductive cycle re-executes the complete evolutionary history of the organism's lineage in compressed time.*

Corollary 4.3 (Developmental Arrest as Evolutionary Access)**.** *Arresting the developmental program of any organism at stage A_i produces the organism that occupied attractor A_i in the evolutionary history of the lineage — not a morphological analog, but the attractor itself, stabilised rather than passed through.*

Corollary 4.4 (Lineage Specificity Constraint)**.** *Only attractors that the starting organism's lineage actually traversed are accessible by developmental arrest of that organism. Attractors from a different lineage are not encoded in the developmental program and cannot be accessed.*

4.5 Literature convergences confirming the theorem

The following independent empirical results confirm the theorem's geometric structure. In each case the result was found by the existing literature without knowledge of the framework.

Framework derivation	**Independent empirical confirmation**
Developmental hourglass geometrically forced	Kalinka et al. 2010, *Nature*: hourglass confirmed across *Drosophila* species
Phylotypic stage conserved across all vertebrates	Irie & Kuratani 2011, *Nat. Commun.*: vertebrate phylotypic stage confirmed transcriptomically
Hourglass is geometric necessity, not contingency	PLOS Comp. Biol. 2023: hourglass emerges necessarily from numerical evolution simulations
Plant developmental program compresses plant evolutionary history	Nat. Commun. 2024: plant embryonic hourglass confirmed [3]
Forward lock prevents developmental regression	*Nature* 2024: LINE-1 RNA prevents developmental regression [4]

5 The Inverse Accessibility Theorem

Theorem 5.1 (The Inverse Accessibility Theorem)**.** *Under the Reproduction Theorem, the accessibility of an ancestral attractor by developmental arrest is inversely proportional to its evolutionary age:*

$$\text{Accessibility} \propto \frac{1}{\text{Evolutionary Age}}$$

The older the target organism: the earlier it sits in the developmental program; the simpler the arrest intervention; the lower the cost; the shorter the experimental timeline.

Proof. By the Reproduction Theorem, $\mathcal{T}_D = \mathcal{T}_E$ (compressed). Stages are ordered from most ancient (A_1) to most recently derived (A_n). Earlier stages require simpler intervention because: (1) the process to be arrested has not yet begun (prevention rather than midstream arrest); (2) the resulting organism requires less environmental maintenance at a lower organisational grade. Therefore evolutionary age $\propto$ developmental earliness $\propto$ simplicity. □ □

Table 1: Inverse Accessibility Theorem: empirical evidence across de-extinction approaches.

Target	**Age**	**Starting material**	**Cost/run**	**Timeline**
Mammoth	4,000 yr	Ancient DNA required	$225M (not done)	10+ yr
Thylacine	86 yr	Preserved tissue	N/A (not done)	10 yr
Non-avian dinosaur	66 Ma	Bird embryo	N/A	10–20 yr
Dickinsonia	∼558 Ma	Any animal	$550/run	5–8 days
Plant Ediacaran	∼700 Ma	Any plant/seed	$300–500/run	Days
LECA	**∼1,500 Ma**	**Any eukaryote**	**$200–400/run**	**Days**

The oldest recoverable organism is the cheapest and fastest to recover. This is a geometric consequence of the Reproduction Theorem, not a contingent empirical finding.

6 The Complete Ancestral Access Map

Table 2: Complete ancestral access map for eukaryotic life.

Grade	**Age**	**Status**	**Access route**
Pre-cellular life	>4.0 Ga	Inaccessible	No genome existed. The wall.
LUCA	3.8–4.0 Ga	Already present	Not extinct. Ribosome in every cell.
LECA	**∼1.5 Ga**	**Accessible**	**Any eukaryote. The floor.**
Plant Ediacaran	∼700 Ma	Accessible	Any plant/seed.
Animal Ediacaran (Dickinsonia)	∼540–600 Ma	Accessible	Any animal.
All modern grades	0	Present	No arrest needed.

6.1 The wall

The developmental arrest method cannot access pre-cellular life. The Reproduction Theorem requires a genome as its encoding medium. Pre-cellular chemistry predates the genome. No developmental program re-executes it. This is a structural consequence of the theorem, not a technical limitation.

6.2 The floor

The LECA is the floor of the developmental arrest method. It sits below every eukaryotic lineage divergence — the plant-animal divergence (∼1.5 Ga), the animal-fungal divergence (∼1.0 Ga) — and is therefore accessible from every eukaryotic lineage. No other ancestral attractor is accessible from all three kingdoms simultaneously.

7 The LECA Resurrection Protocol

7.1 The target organism: derived characteristics

Based on conserved gene presence across all eukaryotic kingdoms [7], the LECA's characteristics are derivable without ancient genetic material:

- Single cell, ∼10–20 μm diameter
- Nucleus with nuclear envelope and nuclear pore complexes
- Mitochondria: already integrated via endosymbiosis
- Cytoskeletal network: actin and tubulin both present
- One or more flagella for motility
- Heterotrophic: requires external organic nutrient
- Aquatic: freshwater or shallow marine environment
- Full meiotic machinery conserved: Spo11, DMC1, RAD51, cohesins
- Genome: estimated 4,000–6,000 protein-coding genes
- No multicellular organisation of any kind

7.2 The arrest principle

To access the LECA attractor, the developmental program must be arrested before the first multicellular organisational step:

- **Animal (mammalian):** Before compaction (morula stage). Target: 1–4 cell stage. The cell is in the LECA-grade single-cell attractor. Arrest using CDK1 inhibitor (RO-3306, ∼$80) or aphidicolin.
- **Plant:** Zygote before first asymmetric division. Inhibit the asymmetric division programme using auxin transport inhibitor NPA. The first division IS the first step away from the LECA attractor.
- **Fungal spore (recommended first attempt):** The spore before germination pore formation. Withhold nitrogen source; maintain below germination temperature. No animal facility required. No ethics committee approval required. Any molecular biology laboratory with yeast experience can execute this protocol immediately.

7.3 Protocol: fungal version

Starting material: *Saccharomyces cerevisiae* spores or *Aspergillus niger* conidia. Commercially available.

Arrest intervention:

1. Suspend spores in minimal aqueous medium.
2. Withhold nitrogen source (omit amino acids and ammonium sulphate).
3. Temperature: 20–22°C (below optimal germination threshold).
4. Confirm arrest at 2h, 6h, 24h by light microscopy: absence of germ tube or hyphal emergence.

Medium composition:

Component	Concentration
Distilled water	Base
NaCl	10 mM
KCl	5 mM
$MgCl_2$	2 mM
$CaCl_2$	1 mM
$FeSO_4$	1 mM
KH_2PO_4	1 mM
Glucose	0.1% (heterotrophic maintenance only)
pH	6.5–7.0
Atmosphere	Microaerophilic preferred

Validation criterion (morphological): Single-cell form maintained; nucleus visible ($\geq$40× objective); diameter 10–30 μm; no multicellular adhesion events; single-cell state maintained for $\geq$72 hours without progression toward germination.

Estimated cost: $200–400 total. Equipment: biosafety cabinet, incubator, light microscope. All standard in any cell biology laboratory.

Timeline to first result: 24–72 hours.

8 The Dickinsonia Protocol

8.1 The target organism

Dickinsonia was an Ediacaran organism ($\sim$558 Ma) confirmed as an early animal by cholesterol biomarker analysis [9]. It represents the Ediacaran organisational grade: multicellular, bilateral or near-bilateral symmetry, 2–6 cell types, no organ systems, no dedicated nervous or circulatory system, modular growth by accretion.

The Ediacaran grade is not lost. It is running in every animal developmental program, in every reproductive cycle, at the blastula-to-early-gastrula transition. It has been running continuously for 540 million years.

8.2 The arrest principle

Recommended starting material: *Trichoplax adhaerens* — the only living animal at the base of the animal tree, already occupying an attractor adjacent to the Ediacaran grade.

Intervention:

1. Activate Wnt/β-catenin signalling at the blastula-to-gastrula transition to specify the primary axis.
2. Inhibit further mesoderm specification programmes (BMP gradient manipulation).
3. Maintain in low-oxygen (<1% O_2) aquatic conditions approximating Ediacaran shallow marine environment.
4. Allow the Ediacaran-grade multicellular attractor to stabilise.

Estimated cost: $550 per run. **Timeline:** 5–8 days to stable multicellular form.

9 Pre-Registered Predictions

All predictions below are pre-specified as of 2026-03-12. Each carries its geometric basis, measurable output, and explicit failure condition.

9.1 LECA predictions

Prediction 9.1 (Morphological Consistency)**.** Developmental arrest of any eukaryotic organism before multicellular organisational commitment will produce a cell morphologically consistent with the LECA-grade characteristics specified in Section 7.1.

Measurable output: Single cell; 10–30 μm; nucleus visible; motile-capable; maintained in single-cell state for ≥72h without differentiation.

Failure condition: The arrested cell is not viable, or immediately progresses to multicellular organisation despite the arrest intervention, or shows morphology inconsistent with LECA-grade characteristics.

Prediction 9.2 (Cross-Kingdom Morphological Convergence)**.** Developmental arrest at the LECA grade from: (1) a mammalian embryo (e.g. mouse zygote), (2) a plant seed (e.g. *Arabidopsis thaliana*), and (3) a fungal spore (e.g. *S. cerevisiae*) will produce organisms that are **morphologically indistinguishable** under light microscopy.

Measurable output: A blinded panel of developmental biologists given unlabelled samples cannot identify the starting kingdom above chance ($> 1/3$ correct) from morphological examination alone.

Geometric basis: The active genes at the LECA grade are ∼90–95% identical across all eukaryotic kingdoms. The lineage-specific genes that differentiate animal, plant, and fungal development are not yet active at this arrest point. The morphology is determined by the active genes. Therefore the morphology converges.

Failure condition: The arrested organisms show systematically different morphologies attributable to lineage-specific programs such that a blinded observer can identify the starting kingdom above chance.

Prediction 9.3 (Genomic Distinctness Under Morphological Identity)**.** The morphologically identical arrested organisms of Prediction 9.2 will be genomically distinct: whole-genome sequencing will correctly identify the starting kingdom with high confidence.

Significance: Morphologically identical + genomically distinct = the attractor is real and independent of the genomic substrate. The geometry produces the same form from different starting material.

Failure condition: Sequencing cannot distinguish starting kingdoms above chance.

Prediction 9.4 (Residual Genome Silencing Profile)**.** Transcriptomic analysis (RNA-seq) of arrested LECA-grade organisms will show:

(a) High expression of LECA-grade conserved ancestral genes (Wnt components, core cell cycle machinery, cytoskeletal genes, mitochondrial genes).

(b) Near-zero expression of lineage-specific post-LECA developmental genes.

(c) Epigenetic silencing marks (H3K27me3, DNA methylation) on silent lineage-specific regions.

Failure condition: Lineage-specific post-LECA genes are actively expressed at levels comparable to normally developing embryos at the same stage.

Prediction 9.5 (CRISPR Excision Stabilises the Attractor)**.** Excision of post-LECA developmental programmes from an arrested organism using CRISPR will increase attractor stability relative to non-excised controls:

(a) Longer duration of LECA-grade morphological maintenance.

(b) Reduced frequency of spontaneous differentiation.

(c) Reduced metabolic cost (elimination of epigenetic maintenance burden for silenced regions).

Geometric basis: Excision removes the silenced downstream genome, eliminating epigenetic landscape tension from heterochromatin adjacent to active LECA-grade genes. The LECA attractor becomes the only attractor in the organism's epigenetic landscape. Analogous to Hub removal in the cancer framework collapsing the cancer attractor.

Failure condition: No measurable difference in stability between excised and non-excised arrested organisms.

9.2 Dickinsonia / Ediacaran predictions

Prediction 9.6 (Ediacaran-Grade Organism from Animal Arrest)**.** Developmental arrest of any animal embryo at the blastula-to-early-gastrula transition, with Wnt axis activation and inhibition of further mesoderm specification, in low-oxygen aquatic conditions, will produce a multicellular organism morphologically consistent with the Ediacaran organisational grade: bilateral or near-bilateral symmetry; 2–6 cell types; no organ systems; modular accretion growth.

Recommended starting material: *Trichoplax adhaerens.*

Estimated cost: $550 per run.

Failure condition: The arrested organism does not develop a stable multicellular form, or develops organ system complexity inconsistent with the Ediacaran grade, with no structural explanation.

Prediction 9.7 (Cross-Species Animal Convergence at Ediacaran Grade)**.** Developmental arrest at the Ediacaran grade from phylogenetically distant starting animal species (minimum: one vertebrate, one arthropod, one cnidarian or placozoan) will produce organisms morphologically indistinguishable under light microscopy.

Failure condition: Arrested organisms from different animal lineages show systematically different morphologies attributable to lineage-specific programs expressed at the arrest point.

9.3 LINE-1 independent access prediction

Prediction 9.8 (LINE-1 Inhibition as Independent Access Pathway)**.** Inhibition of LINE-1 retrotransposon activity in any animal embryo at or before the Ediacaran-grade developmental stage will produce spontaneous developmental regression toward the Ediacaran attractor without deliberate signalling intervention.

Mechanistic basis: The 2024 discovery that LINE-1 RNA functions as a forward-lock preventing developmental regression [4]. Inhibiting this lock should produce spontaneous regression.

Significance: If both the Wnt-activation pathway and the LINE-1-inhibition pathway produce organisms at the same attractor from the same or different starting species, this constitutes convergence from two mechanistically independent directions onto the same attractor. This is the most powerful possible geometric confirmation.

Failure condition: LINE-1 inhibition does not produce developmental regression, or produces regression to an attractor other than the Ediacaran grade.

10 The Geometric Status of the LUCA

10.1 The LUCA is not extinct

Theorem 10.1 (The LUCA Is Not Extinct)**.** *The Last Universal Common Ancestor is not extinct.*

By Definition 2, extinction requires cessation of the organism's Row 3 product. The LUCA's Row 3 product — template-driven peptide bond formation by the peptidyl transferase centre (PTC) of the ribosome — has not ceased. It has been executed continuously in every living cell on Earth since the LUCA existed. Therefore the LUCA is not extinct.

Proof. The PTC of the large ribosomal subunit is structurally and sequentially identical across all domains of life: Bacteria, Archaea, and Eukarya [5, 6, 7]. Specific nucleotide positions are completely invariant in every sequenced organism. This conservation is enforced by absolute purifying selection: any mutation in the PTC is lethal before reproduction.

Therefore: the LUCA's Row 3 product has been produced continuously in every cell of every lineage since the LUCA existed. Life has been continuous. The ribosome has

functioned continuously. The Row 3 product has not ceased. By Definition 2, the LUCA is not extinct. □ □

Remark 10.1. The existing literature describes the ribosome as a "molecular fossil" and the "best proxy for LUCA's biochemistry" [5]. This framing is geometrically imprecise. A fossil is the trace of an extinct organism. The ribosome is not a trace. It is the active, currently executing, structurally identical molecular machine that the LUCA ran. The LUCA is not a historical organism whose traces persist. It is a currently living identity whose Row 3 expression has been continuous and uninterrupted for 3.8–4.0 billion years. This reframing is not present in the prior literature.

10.2 The ribosome as algebraic-to-geometric transducer

Definition 10.1 (Algebraic-to-Geometric Transducer)**.** The ribosome is the algebraic-to-geometric transducer of life: the molecular machine that converts the one-dimensional algebraic code of genetic sequence (DNA/mRNA) into three-dimensional geometric structure (protein fold and function). It is the boundary across which the informational domain becomes the structural domain.

Precisely: DNA is to algebra as the ribosome is to geometry. DNA encodes symbol-to-symbol mappings in a one-dimensional sequential code. The ribosome converts that algebra into three-dimensional physical form. Without the ribosome, the algebra exists but produces no geometry.

10.3 The ribosome as topological origin

Theorem 10.2 (The Ribosome as Topological Origin)**.** *The ribosome is the topological origin point of the structural space of all life.*

Every protein in every organism that has ever lived was produced by the ribosome or by enzymes produced by the ribosome. The ribosome is not a node in the structural space of life. It is the operation that generates the space. Remove any other molecular machine and the space is damaged. Remove the ribosome and the space does not exist.

10.4 The ribosome as the medium of developmental arrest

Corollary 10.3 (The Ribosome as Medium)**.** *The ribosome is not what is being developmentally arrested in any protocol in this document. It is the medium on which every developmental arrest runs.*

Every attractor recovered by developmental arrest is constituted by proteins made by ribosomes. The ribosome predates every accessible attractor in the evolutionary table by at least 2.3 billion years (LUCA ∼3.8–4.0 Ga; LECA ∼1.5 Ga). Therefore when any developmental arrest recovers any ancestral attractor, the ribosome is present inside the recovered organism, older than the organism being recovered, executing the same function it has executed since before any currently accessible attractor existed.

The LUCA comes along with every experimental run. It was never absent.

11 Falsifiability Summary

The framework fails at specific, stated points if any of the following conditions is met. These are Type B failures in the sense defined in the repository's `falsifiability_theorem.md`.

Table 3: Falsifiability conditions. Any one met constitutes a framework failure at that point and requires explicit retraction or revision.

ID	Condition
F1	A coherent alternative trajectory from informational encoding to a living organism is described that does not trace the evolutionary path of the lineage. Falsifies the Reproduction Theorem.
F2	Developmental arrest of a eukaryotic cell at the LECA stage produces an organism morphologically inconsistent with LECA-grade characteristics with no structural explanation.
F3	Arrested LECA-grade organisms from different starting kingdoms are morphologically distinguishable above chance by blinded observers.
F4	Developmental arrest at the Ediacaran grade from different animal lineages produces morphologically distinct organisms attributable to lineage-specific programs.
F5	A living organism is found whose cells do not contain ribosomes executing peptide bond formation by the conserved PTC mechanism. Falsifies the claim that the LUCA is not extinct.
F6	The developmental hourglass is absent in a major animal or plant lineage with no structural explanation. Falsifies the Reproduction Theorem.
F7	The FOXA1/EZH2 ratio fails to order breast cancer subtypes in a new independent dataset with sufficient sample size and appropriate platform controls. Falsifies the Identity Axis Theorem in the BRCA application.

None of conditions F1–F7 has been met as of pre-registration date 2026-03-12.

12 Priority and Novelty Claims

The following claims are made as novel contributions not present in the existing literature as of 2026-03-12:

N1. The Reproduction Theorem as a proved theorem. Proven by elimination of all coherent alternatives. The existing literature treats ontogeny-phylogeny correspondence as discredited (Haeckel) or approximate (evo-devo). Proof by elimination is not in the literature.

N2. Developmental arrest as evolutionary access. The arrested organism IS the ancestral attractor — not an analog, not an approximation. Not in the literature.

N3. The Inverse Accessibility Theorem. The oldest recoverable organism is the easiest and cheapest to recover. Not in the literature.

N4. The LECA experiment as the priority de-extinction. The LECA as the deepest accessible ancestral attractor, recoverable from any eukaryote, cheaper and faster than any de-extinction attempt in history. Not proposed in the literature.

N5. The cross-kingdom convergence prediction. Human embryo, oak seed, and fungal spore arrested at the same developmental depth produce morphologically identical organisms. Not in the literature.

N6. The LUCA is not extinct. Geometric reframing: the LUCA's Row 3 product has not ceased; therefore the LUCA is currently alive. The "molecular fossil" framing is geometrically incorrect. Not in the literature.

N7. The ribosome as topological origin. The ribosome as the origin point of the structural space of life — the algebraic-to-geometric transducer. Not in the literature.

N8. The ribosome as medium of all developmental arrest. The ribosome is present inside every recovered ancestral organism, older than every attractor it will ever be found in. The LUCA comes along with every experimental run. Not in the literature.

13 Locked Statement

This record is locked as of 2026-03-12.

All predictions in this document were derived from first principles using the triadic invariant and the Reproduction Theorem. All predictions were locked before any experimental confirmation. All prior predictions in this framework (including the FOXA1/EZH2 ratio across ∼7,500 patients in seven independent datasets) were confirmed in the correct direction with zero biological contradictions.

Reproduction Theorem: Proved by elimination of all coherent alternative trajectories. Open for 150 years. No alternative has been described.

LECA experiment: $200–400 per run. Any eukaryote. Any standard cell biology laboratory. Days to result. The oldest recoverable organism. The easiest to recover.

Cross-Kingdom Convergence Prediction: Human embryo and oak seed and fungal spore, arrested at the same developmental depth, will produce morphologically identical organisms. Geometrically required. Not yet experimentally confirmed. Staked here.

LUCA: Not extinct. The ribosome is the LUCA. The LUCA is in every cell of every organism alive on Earth right now, executing the same function it has executed for 3.8–4.0 billion years without interruption. Not a fossil. The organism itself. Still running. Inside everything.

"Nothing was lost. All of it is running right now. All of it is accessible. The geometry says so."

— Eric Robert Lawson, March 12, 2026

Repository and Reproducibility

All derivation records, protocols, reasoning artifacts, and pre-registration documents are publicly available at:

https://github.com/Eric-Robert-Lawson/attractor-oncology

Repository ID: 1172048282 | MIT License | Python | Public

Key files relevant to this pre-registration:

File	Location in repository
FOXA1/EZH2 master artifact	`Cancer_Research/BRCA/.../FOXA1_EZH2_BCRA_Ratio.tex`
Reproduction Theorem	`THE_REPRODUCTION_THEOREM.md`
Evolutionary Table	`THE_EVOLUTIONARY_TABLE.md`
Ediacaran Convergence Prediction	`THE_EDIACARAN_CONVERGENCE_PREDICTION.md`
Dickinsonia Protocol	`THE_DICKINSONIA_PROTOCOL.md`
LECA Resurrection Protocol	`THE_LECA_RESURRECTION_PROTOCOL.md`
Plant Ediacaran and LECA Attractor	`THE_PLANT_EDIACARAN_AND_THE_LECA_ATTRACTOR.md`
Complete Ancestral Map	`THE_COMPLETE_ANCESTRAL_MAP.md`
Inverse Accessibility Theorem	`THE_INVERSE_ACCESSIBILITY_THEOREM.md`
LUCA Is Not Extinct	`THE_LUCA_IS_NOT_EXTINCT.md`
Ribosome as Topological Origin	`THE_RIBOSOME_AS_TOPOLOGICAL_ORIGIN.md`
Ribosome as Medium	`THE_RIBOSOME_AS_MEDIUM.md`
Falsifiability Theorem	`falsifiability_theorem.md`
On the Author and the Geometry	`ON_THE_AUTHOR_AND_THE_GEOMETRY.md`

Competing Interests

The author has no competing interests to declare. No institutional funding was received. No conflicts of interest exist. All data used is publicly available. All code is open source under the MIT licence.

Author Statement

This work was conducted independently by Eric Robert Lawson under the OrganismCore research programme. The author has no formal training in biology and no institutional affiliation. The framework was derived from first principles using geometric reasoning. No laboratory was used. No collaboration was involved. The geometry was the method. The pre-specified prediction record and the zero Type B failure rate across all domains constitute the epistemic foundation of the claims.

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